EE5203 L10 Lab Guide

ADC — reading the analog world - Basri Erdogan

Mission: One ADC, two sensors: the knob positions the L08 servo through the 1000–2000 µs contract, and an LDR turns LD2 into a night-light—every reading paced by the Week 7 tick, every number verified in millivolts first.

Prepare before class

- ☐ Re-read the theory slides up to "From counts to control."
- ☐ Bring your working L08 project (TIM2 LD2 PWM + TIM3 servo).
- ☐ Kit: 10 kΩ potentiometer, LDR + 10 kΩ resistor, breadboard, jumpers.
- ☐ Know the answer to: what raw value is 1.65 V on the 12-bit scale?

Learning objectives

By checkoff time, you can:

1. configure ADC1 and justify 12-bit, 84-cycle, single-conversion choices;
2. read two channels by polling, paced by the timer tick—no `HAL_Delay()`;
3. convert counts to millivolts and to control values, all in integers;
4. verify `DR`, `EOC`, and a live `CONT` experiment in the SFR view.

Hardware and files

Item	Required value
Board	Nucleo-F401RE — start from a copy of the L08 project, saved as `L10_ADC`
Knob	10 kΩ pot: ends → 3V3 and GND, wiper → PA0 (A0) = `ADC1_IN0`
Light	10 kΩ resistor → 3V3, LDR → GND, junction → PA1 (A1) = `ADC1_IN1`
ADC	12-bit, single conversion, 84-cycle sampling, clock PCLK2 ÷ 2

Configure in CubeMX

1. Click `PA0` → `ADC1_IN0` and `PA1` → `ADC1_IN1` (both pins go analog).
2. **Analog → ADC1**: Resolution `12 bits`, Continuous `Disabled`, End Of Conversion `single conversion`.
3. Rank 1: Channel 0, Sampling Time `84 Cycles` — one rank only; the code switches channels per read.
4. Generate; find both analog pins in the generated GPIO init.

The channel-switch helper

```
uint16_t adc_read(uint32_t ch) {
    ADC_ChannelConfTypeDef s = {0};
    s.Channel = ch;  s.Rank = 1;
    s.SamplingTime = ADC_SAMPLETIME_84CYCLES;
    HAL_ADC_ConfigChannel(&hadc1, &s);
    HAL_ADC_Start(&hadc1);
    HAL_ADC_PollForConversion(&hadc1, 2);
    uint16_t v = (uint16_t)HAL_ADC_GetValue(&hadc1);
    HAL_ADC_Stop(&hadc1);
    return v;
}
```

Checkpoint A - The knob speaks

Every 10 ms tick flag (your L07 pattern):

```
raw_pot = adc_read(ADC_CHANNEL_0);
mv_pot  = (uint16_t)((raw_pot * 3300u) / 4095u);
```

Evidence for Checkpoint A: the Watch panel shows `raw_pot` tracking the knob over 0…4095; at half turn `mv_pot` ~ 1650, matching your prediction; pause right after a read and find the same number in `ADC1->DR`.

Checkpoint B - The knob commands the servo

```
uint16_t us = 1000 + (uint16_t)((raw_pot * 1000u) / 4095u);  // 1000..2000
__HAL_TIM_SET_COMPARE(&htim3, TIM_CHANNEL_1, us);
```

Evidence for Checkpoint B: a full knob sweep is a full 0–180° sweep, and the pulse can never leave the contract. Predict first: knob at quarter turn → what pulse width, what angle?

Checkpoint C - The night-light

```
raw_ldr = adc_read(ADC_CHANNEL_1);
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, (raw_ldr * 999u) / 4095u);
```

Evidence for Checkpoint C: cover the LDR → LD2 brightens; flashlight → LD2 dims. Explain it from the divider: dark → R_LDR huge → V(PA1) → 3.3 V. Then the SFR experiment: paused, set `CONT` in `ADC1->CR2`, then `SWSTART` — watch `DR` stream by itself.

Individual extension

Pick one, predict first:

- **Hysteresis night-light:** LED fully on below one threshold, off above a higher one. Explain why a single threshold flickers at dusk.
- **Averaging:** mean of 8 readings per tick; compare the jitter of raw vs averaged values in the Watch panel.
- **The chip's own thermometer:** read `ADC_CHANNEL_TEMPSENSOR` (enable the internal channel in CubeMX) and warm the chip with a fingertip.

Acceptance checklist

- ☐ `.ioc`: PA0 and PA1 analog; 12-bit; single conversion; 84-cycle sampling.
- ☐ Reads paced by the tick — no `HAL_Delay()`, no free-running loop.
- ☐ `mv_pot` matches hand math at both ends and at center.
- ☐ Servo obeys the 1000–2000 µs contract from the knob.
- ☐ Night-light direction explained with the divider equation.
- ☐ `CONT` + `SWSTART` experiment shown live in the SFR view.

Individual oral check

- Why divide by 4095 and not 4096 in the map?
- What does the 84-cycle sampling choice buy, and what does it cost?
- Why is the pot's wiper safe to sample, but a 1 MΩ divider is not?
- PA1 reads ~ 3970 in the dark — show the arithmetic that predicts it.
- Where did `raw_pot` live in hardware before your code read it?

Submit

Submit `main.c`, one Watch-panel screenshot showing raw and millivolt values for both sensors, and your predicted-vs-measured table (knob at both ends and center; LDR dark vs bright).

If you are stuck

Walk the chain: pins really analog in the `.ioc`? `adc_read` returns at all? Then: **stuck at 0 or 4095** — wiring: wiper actually on PA0, divider junction actually on PA1; **values sag or jitter** — sampling time too short, or the pin is floating; **PollForConversion times out** — no conversion started (`Start` before `Poll`) or EOC selection left unset; **servo buzzes at the end stops** — commands outside 1000–2000 µs: clamp; **night-light inverted** — your divider has the LDR on top: swap the map or the resistor.